

ActInf GuestStream 082.4 ~ Robert Worden

"Assessment of the Brain Wave Hypothesis"

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hello and welcome everyone this is active inference guest stream number 82.4 on July 29th 20124 We are continuing this series of presentations and discussions with Robert Worden and friends so today we'll be discussing assessment of the brain wave hypothesis thank you Robert and everyone else for joining so please for the presentation okay um this talk is about a hypothesis that we've had previous live streams about the hypothesis is that there is a wave in the brain and this talk is assessing the likelihood of that being true basically and so if you want to see the previous talks which one of which was about the link between the wave and Consciousness then you can follow this link here to an active inference web page so what I'm going to in this talk is firstly try and describe to you what the wave hypothesis is and secondly in point two give some of the evidence which I think points towards there being a wave in the brain and some of the evidence against it uh now the question then arises we've got different lines of evidence how do we combine those lines of evidence to make some overall estimate of How likely it is that there is a wave in the brain and so what I do in point three stage three is at an overall Bayesian assessment of the probability that there is this wave and the answer I come to is about 0.5 is the probability the detailed number doesn't matter what matters is that it's not a tiny probability in other words we can't just dismiss this wave hypothesis out of hand we need to think seriously about what it might mean for the brain If there is a wave and at that point for um we will break for discussion and and feedback from people on the court and after point 4 we'll go on to discuss possible future tests of the wave hypothesis in five and finally after five further discussion again so that's the plan of the talk roughly um and the first part is to describe what the wave hypothesis is so wave hypothesis says that there is some region of the brain which is approximately round in which there are waves going in all three Directions and there's a round region shown with a couple of waves and these waves represent objects in space so the long wavelength wave $\lambda/2$ the green one represents an object at $R/2$ a small distance and the Red Wave with shorter wavelength represents an object at longer distance so

that is the hypothesis that there is this wave region in the brain and neurons couple to the wave as transmitters and receivers so a wave acts as a kind of working memory for a 3D model of space and all the models around the animal in space at the time so that in essence is the wave hypothesis but the important thing about it is how well would a wave based spatial memory perform and you do some Elementary Mass the number of waves you can get in this round region it's approximately if there's some minimum wave length which I call Λ in then the number of waves you can get in the in the space is D over Λ^3 roughly so that's quite a lot of things you can represent in the wave and the spatial Precision you can get is of the order of Λ Min over D and that can be quite small so the benefits of this wave storage are you can get high precision and that I claim you can't get out of neurons you can get high capacity to represent lots of different objects you get low spatial distortion and generally it is a very simple and direct physical representation of 3D space it really acts like a hologram so the question is is there any evidence that there is such a wave in the brain and I think there are three kinds of three main pieces of evidence the first one comes from the mum Thalamus and the second one comes from the central body of the insect brain now the pamus is shown in the left in green for three mammals the point about the thus is it's approximately round so it's well suited to hold a wave and it has all the right connections to hold a representation of space it has all the sense data coming in pretty much all the sense data and it has connections to all the cortical regions that are making sense of that sense data so there are more specific reasons about the phalus but I'll come to those later so that's why the thalamus is a likely can holding the wave and similarly in insects there's a central body of the insect brain which is a small part of the center of the insect's brain which its shape is remarkably preserved across all insect species and it's nearly round so the central body like the phus is well suited to hold the wave and it's connected has all the right neural connections to holder representation of space so those are the first two pieces of evidence now the third line of evidence comes from building neural computational models and it's a thing that I think has not been much remarked upon um 3D spatial com cognition is very fundamental it's the first thing an animal has to do to understand the space around it in order just to move now I've tried building 3D models of space at M David marah's level too and the point is that to build a model of space in the brain it has to be fairly precise a approximately 1% and you can see this because vision and other sense data is fairly precise and there be no point in getting that precise sense data if you didn't make a a fairly precise model of space out of it however it's very difficult to do that with neurons because in a fifth of the second which is approximately the time an animal has to have

to construct his model of space a typical neuron Spike train has only about 10 spikes in it and so this can represent a real number such as a dimension in space with a Precision of approximately square root one in square root 10 parts in other words 1 and three or 30% and that Precision is very far out from the Precision which I think you need to represent 3D space I think why this has not been recognized or not been spotted is that we've done neural modeling of the brain for years and years but everybody models neurons as highly precise reproducible uh units with outputs of real variables as is shown in the top of this figure the output of a neural neuron is usually model as a real variable with very high Precision so you can get high spatial Precision out of that whereas real neurons they have Spike trains and in a short time a spike train doesn't represent a number precisely at all so this I think has not been spotted uh and I think it makes a huge gap in neural models of the brain and as I've just said the wave storage can get much better Precision so wave storage might fill this Gap and that is the third piece of evidence for a wave in the brain so in summary three pieces of three main pieces of evidence beamus the insect central body and these neural computational models which haven't model model 3D space properly and I think that's a bit of an embarrassment for the neural model of the brain really so uh if you have those pieces of evidence and some evidence against the wave which I'll come to uh how do you assess the overall probability that the wave is there well it's a binary hypothesis either there is a wave or there's not a wave and we want to estimate the probability that there is one based on several pieces of evidence and this in analogy is like a biased coin in other words How likely is that coin going to turn up heads on the next time we toss it and you do various experiments one experiment is toss the coin M times and you get heads end times so what you do is several different experiments you might toss with your left hand might toss the coin from an airplane and you might build a special coin tossing machine and with each one you do some experiments and so the best estimate coming out of all those experiments is the sum of all the heads over the sum of all the coin tosses and so this is the way you the basian way that you combine the evidence from different lines of evidence on the same hypothesis you a lot a weight to each one which has a certain subjective element to it and then you sum them all according to their weights and that's what I'm going to do to make a basin combination of the different lines of evidence about the wave hypothesis and I just what you do is you simply you a lot coin tosses to each line of evidence and you SU the coin toses together that I think is uh how people do science roughly and it is if you like it's in the textbooks as the basian philosophy of science so when we do this assessment what answer do we get and I've summarized the answers in this little table and I've

put red rows for things that are mainly against the wave hypothesis and green rows for things that are mainly for it and the two red rows you'll find are prior considerations our prior prejudices and the idea and biophysical mechanisms and I've given those large negative weights I've done that deliberately to show that the overall result is not too sensitive to those negative weights you can adjust them a bit and you won't change the result much and the positive lines of evidence are the three things I talked about the thalamus the insect central body and computational models and a couple of others there is some evidence from phenomenal Consciousness and there are some other considerations so what I'm going to do quite briefly is go through those different rows in the table summarizing basically what the considerations are I won't spend too long on them because the tables are not that interesting but you can come back to anyone or you can stop me on anyone ask questions if you like or come back to anyone in discussion so so what is the prior considerations that go against the wave hypothesis first one I've put in a big weight of 20 for oam's razor in other words it's an extra hypothesis we just don't want an extra hypothesis so let's put a weight against it and so I've put a big weight of 20 against that it's bigger than I think is realistic but I've done so purely to make it clear that the eventual answer is quite robust against that and there's a couple of other arguments I've put in which people might make one is the classical neural model of the brain has served as well so why do we need anything else but I think that's a weak line of argument because the classical neural model of the brain hasn't worked that well that we're ready to that we've described all the brain there's still massive number of things we don't know about the brain so that's a weak argument and the second argument is well we never seen a wave so surely that means they're not one but that that again as a weak argument I think because we just don't know how to look for this wave and it probably has very low intensive reasons I'll come to later so those are weak arguments but nevertheless I come up with 25 points against the wave 25 Point against the wave from prior considerations so now we move on to positive in evidence for the wave and the first one of these is the mamalian thalamus there is both General evidence which I've talked about in other words the phalus has the right shape and it's in the right connect has the right connections to all the three 3D model ens and there's what I call Smoking Gun gun evidence something that's much more specific about the thalamus which points to their being a waven now I'm going to come to that later in the talk so I won't go into that detail in detail now but in summary I think Thalamus offers quite a lot of positive support 14 against two for the wave then we come on to the insect central body there is again the general evidence that the insect central body has the right shape and it's highly

conserved across all insect species now I think that is quite significant in other words every insect in the world has the same shape pretty much the same shape of its central body so the shape must mean something whereas you compare that with other parts of the insect brain like the mushroom bodies for instance they have very variable shapes and they just have to make neural connections but this is a clue that the central body of the insect brain is doing something other than making neural connections and the idea of a wave in it fits it quite well um and I think there's other evidence from the from the insect brain mainly that the insect brain has very few neurons in it so there are probably not any neurons to make a decent representation of 3D space but we know that insects do it rather well so that's the central body next is the computational modeling as I said 3D spatial cognition is a vital function of the brain but that as far as I know Carl can put me right on this there is no working neural computational model of it especially when you ask for a model Which models Spike trains rather than idealized theoretical neurons and in any subsec time intervals neurons give a big error rate about 30% whereas we really need 1% or something to construct decent spatial model so the evidence yeah may I ask just a Precision here the the point here I think if I follow you well that we should all bear in mind is the model that is needed performance in real time so what you are saying is the lack of precision make come also because of some time pressure that for the to be valid should be able to build and update this 3D model of space in quote unquote real time correct yeah very short time when you need that model of space to be really up to date so you only have a in mammals you have a third of a second and insects you have even less time and in that time neurons real neurons are very imprecise that's the argument okay so that's the three lines of evidence for the wave um one line of evidence I think that goes against is biophysical mechanisms in other words how does this wave work and as far as I know no biophysical model for the wave has ever been proposed and we make again the rather pessimistic assumption that the chances that we'll find one are small will come to that later but we've put 10 points against the chances of finding a decent biophysical model of the wave now I think that is very pessimistic especially in view of the the point that evolution is clever than we are and so even if we can't think of it Evolution may be able to think of it and there in in this table there's some other biophysics evidence which approximately balances out so the biophysics evidence is taken to go against the wave broadly uh uh finally the last thing I'll consider is phenomenal Consciousness and this was on the last live stream that if there is a wave in the brain it leads to a new theory of how Consciousness works and you can see that on on that live stream and this theory is I think is quite nice it solves a lot of the problems of

neural theories of Consciousness but we don't want to put too much weight on it because Consciousness theories are so controversial but I think what we can do is that specify Consciousness the the conscience that we have right now of the space around us must come from something happening in brain and our spatial Consciousness is quite detailed and precise and so that tells us however it's done that the model of space in the in the human brain is quite detailed and precise and so that again if you come back to the point neurons can't do it but way might be able to do it that is further evidence in favor of the wave but again because Consciousness is controversial I haven't given it big weight at all I've given it four points so when we and there's some other evidence that I W don't think I'll go into so when we combine all the all the evidence here's the scores we've been keeping been adding up here's the coin tosses which came out heads here's the coin tosses which came out tails and we have a total of 91 coin tosses of which 47 came out heads so the basically we were deliberately pessimistic in stacking up the negative evidence against the wave but in spite of that the positive evidence is one and quite literally the many pieces of positive evidence add up and so we get to probability 0.52 or if we take a simpler view that we just give every line every piece of evidence one vote we get 31 votes and 24 of those are in favor of the way so we get a more positive number out of that and my main point is not the number 0.5 or whatever it is but the the main point is it would take very strong NE negative evidence to change this and I don't know of that evidence and so my main point is really you can't easily dismiss this wave hypothesis and if you can't dismiss it then it raises some big questions I think if the wave exists we need to really rethink how the brain works it's a very different model of the brain we have um and if not by the way there's still a huge unsolved problem because how do these imprecise neurons do 3D spatial cognition so um um what I'm doing the trouble with the one trouble with the wave hypothesis is it really rests on the work of one person at the moment that's just me that's been interested in it but science requires reproducibility and debate between lots of people so what I'm asking people to do watching this live stream and asking you to do is please look at these ideas and assess them for yourselves uh please provide some commentary of your own on the wave whether it's Specialized or generalized or whatever please challenge the ideas please get a hold of the spreadsheet you can download the spreadsheet and and change the way Waits and see what you end up with or please bring new evidence to Bear there may be some evidence against the wave for it that I don't know about and please agree or disagree and what I'd like to do is to collect that commentary and to make that commentary available when people want uh on this website you can get the spreadsheet

and please therefore send your comments to my email so that's the call for commentary and we come on now oh by the way this is very important Carl Friston has endorsed the call for commentary I hope you're on the call Carl um based on this short description of the wave which I'll leave you to read Carl writes this is the Bayesian waves in a brain hypothesis and it challenges conventional views of belief updating in the Bayesian brain and it offers a perspective on neuromorphic computation that's supported not only by theoretical accounts of functional brain architecture but receiving empirical evidence that speaks to a fatigue mechanisms but the important part to me is it would he says at the end it would be very useful to see other people's take on this proposal for Mortal computation in the brain so please send in your comments but on the next slide just before discussion um a few words of explanation of mortal computation this was introduced by Jeff Hinton in 2022 mainly in the context of neural net learning and the paper I'm mainly referring to is by Alex and Carl in 2023 where they relate mortal computation particularly to Markov blankets but also to a lot of other things there are so many ideas in that paper I can't possibly summarize them but basically these are proposals for brain inspired my biometric Computing I'll summarize I'll a quote from that paper summarizes what mortal Computing means to Alex car software is Immortal because it can be copied to different hardware and still be executable mortal computation means that once the hardware medium upon which a program is implemented fails or dies the knowledge behavior and functionality will cease to exist so that's model computation um my take on it is that direct physical computation which is the kind of computation the wave does is simpler and more efficient than a universal Computing machine like a universal Turing machine or a universal Von Neuman computer and in particular Immortal software needs very reproducible Hardware but that doesn't work out in the brain for two reasons two big reasons firstly no two neurons are identical in the brain I believe and second secondly even one neuron never does the same thing twice so coming on to the relationship between wave-based cognition and Mortal computation I think wave based spatial competition is mortal and it's morphic to use a word from this paper in other words it depends on the physical shape of the thing holding a wave in terms of their 5es model it is Elemental because I think spatial cognition is a very basic thing that any animal needs it's embodied because embodied in space we're all embodied in space and it is inactive because we have to move in space in order to understand 3D space so wave computation is tied to a visible wave and I will claim that we need to understand 3D space before we need to learn things and that's why way spatial cognition is really very Elemental and an interesting consequence of this if you believe there's a wave in the brain and if the the wave is the source

of Consciousness then you get the conclusion that Consciousness is Mortal I think we knew that but I think it's quite interesting so onto discussion and um these are some of the points I would like to hear about in discussion perhaps you can take them in any order but um Dan do you want to take over thank you steering the discussion yeah how about Alex if you'd like to give a first comment and then anyone else can start to jump in can you hear me okay let me know if the background noise is really bad because I'm in like a cafe so first of all I think this is very interesting I did read Robert uh your call for commentary paper uh before as well I did have a few notes I mean there's a lot of things that we could talk about but I I do really appreciate the connections to mortal computation thesis uh which is a corollary of the free energy principle so I think this is really nice and uh I did I did like the argument about that waves are morphic thus you know that they fit in the framework because we kind of need that uh otherwise we're just dealing with Immortal computation so I had actually some questions I'll just throw at you that might they might even just be for further information so one one comment question is I don't know if you're aware of Robert West at uh Carleton University in Canada uh who has worked on holographic declarative memory uh one of your comments earlier about uh things being holographic and are there neural models that try to explain or try to model things that are wavelike but these aren't really wavelike uh holographic declarative memory comes into play where they use complex numbers and complex number representations which complex numbers are often used to model waves um so I was curious if you had known about that work or if you did uh why would that be insufficient or how that might play in the framework and I'll lump with this another body of works that way I don't break this up into too many questions uh resonator networks or resonating fire neurons have a wave like construction about them uh uh what is his name Eugene zovich in computational neuroscience proposed uh the zovich neuronal model which is a very complex spiking neural model and those Dynamics can be uh constructed such that they emulate resonate in fire Dynamics or wave like Dynamics and I was wondering how that if you were aware of that if not that might be very relevant in a potential neural model that could be a step in the right direction since you said that we don't really have the neuronal models or neural computational models that get at this notion of wave like comp computation um they might not be completely relevant so I was kind of curious about that I'll stop there because then I have like one or two other questions so I was curious if you had any comments on that thank thanks a lot for that the quick answer is no I'm not familiar with either of those and I will go and look at them as soon as I can uh they may well have insights uh what I'd say about the wave hypothesis is it is

the key thing about the wave and I didn't emphasize this enough is it's got to be a working memory so it's got to hold information for a fraction of a second and that makes it rather different from anything that is Neuron oriented but I will certainly read those to with great interest thanks a lot yeah I'll try to uh send Daniel or you some links to some of those papers to point you into those directions and then I guess I'll um ask my other question uh I mean I can also lump in chrysis work on spawn because there are lots of neuronal models that try to get at modeling the Dynamics through spikes and again maybe this goes into that spikes just don't give us enough Precision because in the paper I didn't quite see that but in your talk I did um which was why why do you claim that we don't model spikes well as we have a long history of spiking neuronal models and computational Neuroscience and brain inspired computing that do work very well now again I don't want to claim that there this is where Carl could chime in as you mentioned him earlier uh if there was any that have been used to model aspects of complex spatial cognition that I think might be lacking but like we have Huxley model again I mentioned the isovic model uh fiton nagumo there's a lot of these complex Dynamics uh that can actually be used to engage in very complex machine intelligence tasks uh that for example you can train simple robotic Control Systems uh with like modulated forms of Spike timing dependent plasticity which is also a very rich and long history in Neuroscience for plasticity of course and this is where we're now we encoding in synapses I'm not sure that this feels at odds uh with the wave hypothesis but rather could be tools uh that you could be used to build wav like models and again I was just curious to know if there's still something about those particular spiking neuronal dynamics of which we have decades and Decades of work that are just lacking to give you what you seek um and again is it just because we can't really represent things with 10 spikes in five milliseconds that even if we represent them Faithfully to the neurophysiological data uh that's still insufficient for representing what you want and you're just saying we need another mechanism that's the part where I was a little confused yeah I I think I please send the references I love to see those papers I think specifically what I'm saying is in I haven't seen a detailed 3D model of space to the level of resolution like a B's eyes or our eyes that's actually built with neurons firing there may be lots of uh neural models built with Spike trains I'm not aware of like I'm not aware of any models within the Fe active inference work I'm not aware any models outside like Edmund rolls biset or um Jeff Hawkins stuff and so on I believe that all of those use precise theoretical abstract neurons okay that's fair um I think uh yeah I think I I think I get what you're trying to say I do I do feel like there might be some interesting work out there like for

example I'm sure you've heard of the recent model computational models in the OS for like grid cells which is used as an important part of oh yeah no I think looking at grid cells is an important thing we have to do okay that might be complementary rather than an opposition as I I do like the way hypothesis I don't necessarily think that this is data that would necessarily refute it and this is where Carl might chime in uh because again if he's aware of any good neuronal models or computational neuron models that actually get at this that I'm not sure that great thanks Carl or Kenard jid feel free perhaps I'll take the lead and just um respond to those those comments I think um you could argue both ways um but it does rest very sensitively on whether the wave is a description of neuronal Dynamics under the neuron Doctrine or whether the wave hypothesis commits to some divorce between the wave however instantiated and the uh neuronal activity so certainly um the the if you say that the wave um is an appropriate description of Ensemble Dynamics um on the cortex or in the hippocampus or in the thalamus um then one could easily argue that um it will be difficult to disprove the wave of hypothesis and indeed there are people's uh whose entire career is built about uh built upon understanding um spatial cognition and Beyond in terms of cortical field theory for example so we just look at the uh oh there are many examples here but the sort of U classical example that comes to mind is the work of Peter Robinson Australia um and developing quite sophisticated wave likee models of um neuronal Dynamics idealized can I come in there car ju just briefly I mean the key thing about this wave hypothesis it's got be working memory so it's got to store information for um and and it is a separate non-neural wave that's basically the way it's pitched and it has to be a wave which can store information for a fifth of a second or a third of a second or something like that and so it's not just electric Fields passively caused by neurons yes so that speaks to you know what what does the the wave hypothesis um commit to uh and you're giving a very clear answer that uh it is not merely articulating um Ensemble dynamics that will could transcend neurons we could be talking about G cells we could be talking about Quantum coherence you know but the the key thing is that you're summarizing spatial extended Collective dynamics of active matter in the brain um yeah in terms of uh INF frequency space basically so you could you could say what this is exactly the same move that quantum physics makes it's instead of yes absolutely yeah um you basic I mean at its simplest you can regard quantum physics as basically articulating Dynamics um be specific about this sorry articulating density probability density Dynamics in frequency Space by appeal to complex numbers and for transforms um and um what's what's the benefit of doing that well you can use foro transforms and all the orthogonalization and all the linearization of various

problems that ensues um what's the difficulty well the slight problem with um modeling probability waves is that once you take a Fourier transform there is no natural way to uh impose a symmetry constraint that you would need in order to interpret as a probability so what do you do you have to invoke a Born Rule and then think about probability amplitudes but the message here is that there is a great utility in taking any arbitrary Dynamics in some Metric space and specifically density Dynamics and projecting them into the space of waves um and from that point of view you know one could argue that this the wave hypothesis is not a hypothesis it's actually a principle or a proposal for a better way of understanding Collective Dynamics and in that sense I think there are lots of examples of that and I come back to cortical field theories Quantum field theories of um um applied to um sort of the brain um that have the right kind of dynamics will have exactly the right kind of um um working memory so I'll just give you a couple of examples there is work um looking at um Chimera States in cortical field Theory um where um one can find things like solitons which are you know brings us back to last year's conversation condensates and the like more interestingly more recently um there's been empirical evidence from both EEG and fMRI of vortices on the cortex um which are if you you know one way of understanding um classical like Dynamics um in the space of waves that have the kind of itinerancy and um solenoidal aspects that do remember things by returning to the same state um so if you know I think there's lots of empirical evidence U that and thereby um meaning that they have memory in the sense of say a Konari map um and these are explicitly articulated in terms of traveling waves that have this circular noal aspect that can be empirically detected however so I you I think your answer the question is there any empirical evidence for a formulation of um brain Dynamics Beyond um a commitment to particular the neuron Doctrine um that can be articulated in the space of waves that fulfills the the uh requirements of precision for memory and representation I would say yes but that does not answer the question is are these waves mere descriptions or of one or many kinds of descriptions of some underlying um some underlying Dynamics um or does the spatial wave uh the wave hypothesis of the brain say that the actual substrate the Mortal substrate cannot be just limited to uh message passing among uh among neurons and that's what I meant by the neuron doctrine that I think is yeah yeah that I think that is my view it cannot be reduced to neurons basically yeah that's what I understood from your previous talk in the slide you shared before of course there is there are a lot of waves in the brain for instance related to electromagnetic induction and there are a lot of old and very powerful models go back to nun years you know to connect with SCS discussion of EG Dynamics but I think one of your

points is that the wave whatever that is uh cannot be just an epiphenomenon and it has to have an inductive power on neuronal Dynamics in return um there is some model I think that are emerging uh or some theories that are emerging about electromagnetic induction uh uh like electromagnetic wave in the brain that with inductive power on current in the in thees um I haven't followed that literature for a while as I told you before I give up gave up on Neuroscience uh but um so what you require is really something that has a you know bottom up and top down quote unquote sort of mechanism to begin with this wave has to be sustained and it has to be in in of itself a form of computation that can then feed back to the neural network that's question yeah Ken would you like to give any comments uh just to maybe a one for everybody a question for everybody might it reveal some of my ignorance here but um is there any sort of General results about the capacities of waves for computation relative to other types of computation it's it's a kind of analog in computation isn't it it is analog is very appropriate term for it but I I would say the the role of the wave is more information storage than computation but it it actually may be involved in both but the key thing is it's got to neurons May compute 3D space there got to be some way of storing it with adequate precision and that's what the wave is supposed to do but I should add if I may that Computing 3D space in particular in an active in C kind of a realistic biological uh context for you know animals that's not just representing space it's also acting upon it through Transformations that are not just about change of coordinate for instance under hypothesis there is the projective uh group that is acting on a projective space but then you have the space that is itself filled with features that themselves as their own operators like different groups for and spaces colors and how all these things like perhaps through models of fiber bundle interact is very key so the the computation of um space as you think of it and I think is perfectly correct and and talks to me uh is much more than just uh you know represented representing three coordinate at one point in time I agree I agree just to speak to um Ken's question about are there any laws that would be applicable under the wave hypothesis just going back to the analogy with quantum mechanics if one thinks of the wave as a probabilistic um representation or memory um then you'd certainly be looking to Heisenberg and certainty principle and the de Broglie relationships so I perhaps this could be a question for Robert you know H how would you unpack the spatial Precision in relation to the temporal Precision because you talk about memory but the the uncertainty principle would mean that you can um you can have a certain um spatial Precision at the expense of a temporal precision and vice versa indeed that is one if you like gift of a wave hypotheses certainly as read from the point of view of Elementary uh

qu Quant quantum theory um the the other so question i' reflect to you in terms of the deoy relationship what what what what is the analogy of say energy and mass in in these in these ways you know how would you in how would you interpret this to a sort of 101 physics student um when it comes to with great difficulty um I let me first speak to the spatial Precision I think the uncertainty principle is basically a property of the fer transform a wave that's confined in a certain distance has a certain uncertainty in its wavelength and wavelength is momentum in quantum mechanics and and position is position so that's the UN that's the the origin of Heisenberg uncertainty principle is simply the Freer transform of a bounded wave that speaks to the spatial Precision but as to the actual biophysics of the wave and there is some relationship between wave vector and frequency I don't know what that relationship is but the the gr as you recall uh which way does it go frequency over wavelength is Phase velocity and the derivative of wavelength with respect to frequency is the group velocity now all those questions about what the group velocity is what the rest mass of these if there are Quasi particles what's their rest Mass I don't know but uh for instance one of the more far out hypotheses I consider is as we discussed it last time was that the W is actually held in a Bose Einstein condensate and in a Bose Einstein condensate there are particles called roons that haven't phonons normally have zero rest Mass but there are particles uh called Rons and others which have nonzero rest M and that may be something to do with their persistence in time but nonzero rest Mass gets you into the whole question of what's the frequencies of this wave what are the wave vectors and as far as I'm concerned it's all the Wild West it's all to play for really I I I'm only making suggestions at the moment thanks I'll make a few short comments so thank you all for the discussion and Robert also for making the basian epistemology very clear I think it's really cool also that was very insightful discussion on one possibility the wave is epip phenomenal like David said plays more of like a memory possibly even a memory with no causal efficacy no I'm sorry I did not mean that I mean according to what I understand and is necessary for Rob model the wave must not be a p phenomenal yeah and that's right what we know about EG except some models for instance now there's some models of electromagnetic propagation with notion of feedback induction that should be looked into but overall the way we model EG for instance from the neuronal current is EP phenomenal it's just you know summation at the level of an electrode and so the wave itself is an epiphenomenon and the the model of Robert does not want that I totally that agree and that that's where I was going was sorry to attribute it to you that way but it was yes to to a more active computational role kind of like this Reservoir Computing area um also a general Point kind of summed up by

figure 4.3 in the 2022 textbook is there's like continuous and there's discrete models in space and in time so given the continuity of space and time just at a first pass a continuous representation or model like a wave may have certain advantages over holding it in a discrete way and then um in the act of inference does not contradict folks' work there's the continuous time model on the lower scale and then a discrete more switching slower model that could be like an intentional context oh so so there is precedent in terms of having a lower level continuous model higher level discrete models and uh then more to taking that to the bigger questions there's also some natural empirical correlations with bigger slower things and faster smaller things so to the discussion of well multiple things are represented like space and time some of that is doesn't need to be an open-ended representation it can be like evolutionarily tuned or developmentally tuned or learned um and then last comment just on the insects it was a great point a line of evidence that's very promising because there's a lot of neurophysiology and neuroanatomy available in insects and just to say that the visual region of the the brain this is arthropods and insects and the old factory as well as the mushroom body have more like stereotypical units and they do have more change in size and more variation in evolutionary whereas some of the interesting evidence from what you presented from the insect brain is that the actual geometric volume and size of the central body is relatively conserved even as other parts of the brain have a lot of allometry so that suggests that one way or another either it doesn't need to be explained in terms of fulfilling an adaptive function it could just be oh it's constrained or something like this but if it is going to be playing a role which is a fair assumption in an energy limited system then part of its role can't necessarily be improved by making it bigger or smaller yeah I agree I agree you mean relatively bigger or smaller right because I mean it the central body does scale a little bit with with the insect brain size but it scales less rapidly than the total insect brain size the interesting hypothesis see whether it scales with the visual acuity of the insect yeah how the different regions size and other attributes this could be in mammal and/or arthropods power the empirical psychophysics and the just noticeable differences described or explained by like for example just geometric volume of the old factory L whereas if there is some other variance to be explained yeah well but there will be a lot of correlations between you know neuronal population densities and volume so if volume increase you may have more neurons uh obviously you know that's basic neuro psychology 101 and vision Neuroscience one you know if you lose the neurons your acuity diminishes so there seems to be a strong correlation between your angular precision or acuity for many other things and the number of

of neurons and connections that those neuron makes but after a while if you lose twoo many neurons you have a de differentiations like think about neurod degenerative diseases like you know take alimer for instance where everything get disconnected and there is not enough neurons to represent stuff so that I it would be hard in the C you know given what is fairly know in Neuroscience to reject the importance of the I'm simplifying here the number of neurons the number of neuronal units in your brain and your uh resolutions capacities thank you yeah how about Robert continue with uh part two should I I bash on yeah great well thanks everybody for that uh so here's what we've got to on the road map we've described the hypothesis evidence for thre against we've done the assessment and we're now on to Future tests how can how can we test this way hypothesis and I'll divide this roughly into what can we tell from aonomus from the insect central body from computational models from biophysics and from other sources of evidence so I'll just go through some of those um so um this probability 0.5 is an unsatisfactory place to be we really want to drive the probability up or down and so the first thing to do is to stress test the three main groups of evidence that we have already and then we look at other evidence for and against and there are a lot of possible tests described in this paper which I've given the link to the P tests are all um in this table here now I'm not going to go through all the table uh I'll just say the paper allows you to sample the particular area that may interest you so you can just pick out the section that speaks to whatever test you're most interested in and I'll just do some dipping in sampling myself so the first first thing is the so-called What I Call The Smoking Gun the specific evidence in the thalamus and what this specific ele evidence is is there's a picture of Thalamus here on the right and so Thalamus is a round thing at the center of the cortex which is like a hemisphere around it and what you have to notice is the thalamus in mammals has lots of different nuclei thalamic nuclei and the thalamic nuclei are only very weakly connected to each other but they are strongly connected both ways to the cortex so the question is why don't the thalamic nuclei move out of the thalamus towards the cortex they're going away from each other there only weak connection with other nuclei they're getting closer to the cortex so they're reducing the length of mated axons and that would save a lot of energy so that's the problem with athus under the nuclear model if the nuclei exists just to have neural connections to each other they're very weak neural connections and energy could be saved by moving out each nucleus towards the cortex whereas if the thus has a wave in it there's a good reason for all the nuclei clustering together the nuclei have to be together in the same wave now I've tested this energy argument by actually uh modeling what happens when you move theic nuclear and

approximately I obviously it was very approximate test on the human Thalamus and the answer was yes you can save energy that way so there is a mystery of why the thalamus stays together but that was an approximate test I think now that test could be done much more precisely using connectome data brain connectome data so you could use brain connectome data you could have a computer program that tries out various moves of nuclei to see if they save energy so that that is the test that you could make this argument about the Smoking Gun much more precise and there's a similar test for the thalamic particular nucleus which I won't go into I would say sorry sorry to interrupt I would have kind of is it possible to interact Slide by slide or do you prefer that we yeah absolutely yeah so I I get your uh energy argument in term of neuro development or neuro evolution of anatomy and you know anatomical relationships though I would argue I mean I would not argue I would you know ask you but you know what if in this uh energy model you take into consideration all the inputs and the outputs uh from Sensory neurons for instance from sensory organs the eyes for through the optic nerve uh and of course then the output toward the cortex but also the you know motor input uh I'm not sure uh that if you think about all this wiring then your energy model not forgetting all the white matter that you have you know and connectivity that you have between the cortex and the thalamus not sure that the energy model would fair that well that's the course um well the basic point is that when I did the approximate calculations I did take account of the upcoming you know the sensory neurons coming into Thalamus the point is if the thalamus was is just a relay pushing data on from sensory data to Cortex then moving the thalamic nuclei would be neutral whereas because there's feedback circuit connections you actually gain an energy by moving out towards cortex so this calculation has been done albeit it very approximately and I think the opportunity is to do it much more precisely taking account of all thalamic connections from connectome data uh but you you know of course and that's really a bit uh old for me I haven't been into this literature for a long time but there are B models connectivity it's yeah thalamic nuclei kind of lives in CEO but they actually have Silo I don't know you know there mod but they actually have basal or triad uh uh interactions that are kind of fine some of them are modulated by the reticular nucleus is that with each other I understood from Maravay Sherman there's a quote from him that says there are no connections he puts it very strongly when was that published remind me um well I mean he publishes book about Thalamus in 2006 but yeah since then he's written a lot of papers about Thalamus and and he's well yeah sometime people are very assertive but you have a great all that neuroanatomist that I've met in the past like G van den Pol uh uh they look at the brain in very very very very small in minute

details you know they they do real microscopic Anatomy and the way they see the brain universe is quite different from uh I don't know if this nor anatomist you are talking about work with these methods uh I would be very prudent with connectomic data I think they are useless but anyway that's another so you suspicious of connectome data yeah hold the DWI I mean DTI data and function okay that's my personal opinion but I'm radical I'm almost you know I think they are useless lot I think they are useless but anyway we can have another discussion about that yeah anyway to me there's quite a close analogy between this argument about the thalamus staying together and the argument from Galaxies for dark matter in other words when people looked at galaxies a long time ago they found there wasn't enough mass in stars to hold the Galaxy together by its gravity so they invented Dark Matter so I think when we look at Thalamus there's not enough neural Connections in the thalamus to hold it together against the pull outwards towards the cortex and and so the wave is a bit like dark matter in the thalamus so that's the analogy anyway but I think the connectome test should be a precise way of uh stress testing this hypothesis right the in central body um as I said the gross anatomy suggests a wave but you can't make the same Smoking Gun argument but I think there's a lot of detailed Anatomy that you look at of detailed neural connectivity where the synapses are distributed and where what happens in the compound synapses I think they're very suggestive and there may be not a simple connectome test using nuclei because I don't think there are nuclei in the in central body but there may be some connectome test that can reveal whether the Neuron model actually is sufficient to account for the shape of the central body so this I think is a very open area for investigation I personally like it because I think science should be done on simple things and there's this quote from Ramon kahal that said the melan brain is like a grandfather clock the insect brain is like a Swiss watch I like that so I think we should look at them so the Third argument we should stress test is neural computational models uh I make this claim that understanding 3D space is very Elemental it comes before learning I think because animals have to do that before they can start learning and I've built a model of 3D spatial cognition at level two and that needs High precision and had this argument before now or Spike trains only give much less Precision in short time so what we need to do I think is to stress test this argument by really trying to build a model of a neural model of spatial computation using mortal fibre neural neurons and one good framework to do this is the free energy principle because it's got well established neural process model and if we can make this work with neurons then maybe a wave is not necessary but if we can't make it work with neurons then that that strengthens the case for the wave just another slide on this I

think we should try and build two models say in the F Fe um perhaps using the level two model as a starting point but in the moral neural model we have to model do we model Spike trains or do we model just a variable neural Precision there must be some neural in prison we model if we model neural in Precision how do we do gradient descent because you'll see I've struck out here if you do naive calculation of gradients in other words you compute a quantity and then you shift some input variable small amount and you compute the quantity again to compute the derivative if there are lots of Errors you don't that doesn't work because the imprecision overwhelms the gradient so there's an issue there about how you do gradient calculation I'm not sure how you tackle that obviously you can compute gradients explicitly but it's a more complex computation than Computing a quantity next question how do you do a gradient descent in space and I think in the FP a gradient of a free energy is a three vector and Precision in the F sense is a tensor probably how do you what coordinate system do you use do you use cartesian coordinates I don't know I mean seems to me cartesian coordinates in the brain is a bit unrealistic so the key question I think is I expect that the naive Precision of neurons from Spike trains is not going to do the job so the question is can we have some clever neural Micro Design which gives us High precision and high speed and the final question is do we use a spatial hierarchy as is common in the F or do we use a Blackboard uh again I would prefer using both but the issue of what are the mark of blankets arises there so that's the issues we' have in a pure neural model in a wave neural model how do we model a wave do we build a physical wave now that suggestion is not quite so stupid um one could build it um in the very early days of computing there was a thing called a Mercury delay line memory which actually used a way for storage in one dimension and we could build something like in three dimensions but I think that would be very expensive not as expensive as building a quantum computer but very expensive anyway so do we simulate a physical wave or do we just simulate the memory parameters that we get so simulate the Precision and the capacity and so on I would prefer to start with a simple simulating of the memory parameters and in other words I'd like to move down from level two to level three in successive stages so that's um the Active Vision in Exploration another area we can look at is if there is a wave in the brain it should show up in genomics and proteomics somewhere and there should be some genes which express only in certain places like in the thalamus or in the insect central body or those genes May the proteins may be particularly affected by anesthetics because on the hypothesis that anesthesia disables spatial cognition disables a wave you might expect a a link between those proteins and anesthetic molecules and we're looking for proteins

which not only sustain the wave but also proteins involved in neural transmitters and receivers which exchange information with neurons and I think that's a promising line of investigation if we find any possible proteins they could help to constrain the biophysics which I think comes next biophysics of the brain is absolutely wide open I have no idea really there are some very preliminary clues and I think most of the ideas are probably wrong but the key thing is the wave needs to store information it must persist for fractions of a second so it's probably not purely electromagnetic uh because as David said that's an EP epiphenomenon activity so could it be a coherent Quantum effect uh the problem about quantum mechanics of course is decoherence removes it very quickly and you have to somehow conquer decoherence there are ways to combat decoherence I'm particularly inclined to be interested in looking at nuclear spins there's a chap called Fisher who's looked at spins of the phosphorus molecule which he claims can have coherence times as long as seconds or even minutes um there is this possibly attractive idea that the wave could be in a Bose Einstein condensate or a Fric condensate um so there all these questions all these possibilities I haven't really any strong direction or other but we have to answer the question a how does the wave persist and B how do neurons couple to it and we might get a clue in that respect from single celled animals we'll come to that in the next slide but I'm saying this is a tremendously difficult problem and just because we can't think of an answer we shouldn't underestimate evolution and I think broadly we need to find some possible mechanism before we can start looking for it experimentally um this is intriguing to me um there are single-celled animals like eroid is I can't pronounce it it's a single cell animal it's kind of vanav and it has complex behavior it hunts and eats crustation eggs and it has something in it which looks suspiciously like a lens of an eye and something which might well be a retina so does it have a lens and a retina if so how does it use that information if it hasn't got any neurons how does it guide its own behavior can there be a brain without any neurons and could that be a precursor for all the brains could it arrive evolve in prean Era and and persisted through the caman era so I think there is something to be learned by looking at these single cell animals may I comment again yeah sure I totally agree with you if you look at paramia obviously they are very sophisticated sensory motor schemes they have some there's some evidence that they can have some memory they have can be conditioned they have potentiation some of them like spir spirostomum if I remember well and obviously the computation quote unquote or the analog analog computation if you prefer is going through molecular networks but works exactly as well as a as a collection of neural animals uh so that's also is telling that computation again quote unquote memory and active inference uh can go

through many type of biological networks uh and so we shouldn't think that because the neurons are the most apparent and they are very very sophisticated and complex that uh there are no other kind of unknown for us mechanism of computations that go through biological network of protein network if you wish that I think it's a very interesting hypothesis and that the multiscale interaction between all those things is also kind of an entire universe that we you know barely understand more gal cells are now we understand better that they have not just a trophic hole but a very sophisticated computational role so anyway uh I think that's that's a very important thing to consider in know for the future of Neuroscience great thanks David yeah so I've really finished my talk um over to discussion again um so some of the topics we might touch on with the connecto ISS or particularly the computational model very interested to hear Carl and Alex on that and insect brain studies and even the biophysics so um over to you please naughty um is preparing his um revealing answers and commentary I'll talk rubbish for a moment um so I I just pick up on the sort of practical issue of the connect term I think I I think um David was showing an understandable allergic um response to connector based upon diffusion weighted Imaging using non-invasive magnetic resonance imaging um I think what you're talking about is this connectivity that could be tracing studies it could be um supplemented with problemistic trap toogy but you know so if you just don't use the word connector I think I think um uh you'll be fine just talk about connectivity studies um which are usually definitively um based upon sort of you know staining and tracing um at a microscopic uh level as opposed to right reconstructions um for me um well you know the tests of the hypothesis um there for me the ones that you've offered um they do um invite rather um if like deflation responses so for example the arguments that a single neuron firing um over a 256 millisecond um period cannot um contain the sufficient statistics required required for spatial temporal Precision of the of the sort required for um spatial navigation at a certain space-time scale you know one simple way of counting that well that's true but if you give me 10 to the three of these spiking neurons and I can now just write down um the um Ensemble average then they do um so the whole Precision argument can be I think almost dissolved without making any iral moves or any tests can I come in on that I I think if the test You' you've had your chance fromer no no no this is no just briefly just briefly the test of the Precision argument is to build a model that works and by the way do it for an insect yeah um so I think we're getting pretty close to that but and in so doing um you actually appeal explicitly to mean field approximation which is exactly modeling the probability distribution of populations or ensembles in the S mechanics sense so you know you know the I I'm reminded now of um

Rodney Douglas's ambition for the EU to build a b uh I think if you look at certain um drone rumber applications of Slam simultaneous localization and mapping in Industry actually out there um you will find that the you know not perhaps to the level of a bumblebee but certainly to sufficient to um clean up the airport floor amidst moving uh passengers that kind of technology is already there um and you know I would imagine in about 3 years you will have um the same kind of Technology um operating at the level of drones and and at that point I think you can say yeah we can do this without without a wave hypothesis um yeah but drones with respect have perfect theoretical accurate neurons they don't have mortal neurons that's yeah the point being that you can simulate a population of mortal neurons such that the random uncertainty averages to zero so this is what is meant by mean field approximation so that all of stochastic and Cal thermodynamics rests upon this so the density Dynamics itself now becomes deterministic you are still modeling something which can be quite noisy with lots of random fluctuations and thermodynamics underneath them but the density dynamics of the kind that quantum physics would appeal to is actually deterministic but I'm not I'm not you know reason I sort of was teasing you said you can't respond because it's our turn now was I don't think I don't think these are important arguments to me the important argument is something which came out in the last discussion and you I think Daniel summarize very nicely is is there any top down causation between the wave and the neurons that are controlling my body and my eyes um and um you know because I could argue that it's all bottom up causation and just rewriting neuronal density Dynamics or population Dynamics in Foria space yeah everybody does that even to the extent of um focusing on the relevant variables as the rate of change of phase which Robert you were you saying that a really interesting interpretation yeah that is the kuramoto model of neurodynamics uh you know this has been around for decades people do work in this space but the assumption is that you are basically appealing to a bottomup causation from some underlying neur mortal computation at level of neurons and then you're just lifting it to a level of description in the in in a in a poror space or in a phase space with the right kind of Dynamics making all the arguments that you have made in favor of the wave hypothesis it seems to me the important thing is a top down causation that destroys theological thing so I'm just trying to think about tests of top down causality of course not easy to do but I'm just wondering you know if I squash your Thalamus would you have different spatial cognition so I preserve the connectivity but I actually changed the shape of your Thalamus things like that I think would be more compelling demonstrations of the top down causation which I think is at the heart of your spatial wave hypothesis that it's not just an

epiphenomenal bottomup consequence it actually matters this wave my neurons have to be immersed in this wave to function properly and it matters where they are and in that mortal sense it matters um that these anatomical constraints afforded that support this wave will have an influence on neurodynamics and subsequent belief updating and movements and sense making and projective geometries um so I don't don't have an answer but I would love look into you at at those things that if you like address the notion of circular causality from the point of view of physics and in particular drill down on what would it be mean to demonstrate empirically top- down causation in the sense or the spirit of George Ellis um that that's what I took from that conversation that exchange I I agree hard problem honestly I think even though Carl you said you're just gonna you know Babble or say rubbish I think that was the most important point because that touches on the things that we even discussed in the in the Mortal computation thesis framework um and that was actually so I don't know if I have the wonderful insightful answer that you claimed I might produce but uh just to speak to what you said and uh and what Robert was discussing I think that is what were lies the most important if we're going to build computational models which obviously appeal to me quite a bit uh in the moral computation thesis said well okay the real way to do this is we would take actual substrate like for example a neuromorphic chip if we were doing artificial or we would work with xenobots or anthr robots and build you know little systems of these actual biotic or chimeric type of systems but we did discuss virtual morphology so I think you could just go down to the simulation route which touches on what you said Robert about well we could build waves we have examples of those but those come with some difficulties we could just simulate them and I think the answer to answering the question how do neurons couple to the waves and what does that look like and how do we build those computational might be one of the most important questions I think in building any testable computational model and any useful simulation I do think probably what I was saying in the earlier discussion about looking at resonator neurons or resonating Dynamics uh which has some history and as well as the cognitive math learner and the stuff I've talk about holographic memory which kind of Builds on the complex number representation and the wave representation and how do you encode that in spiking Dynamics I think that might be an interesting and useful building block and I will send Daniel and you some information and then you could look and see if that's rubbish or if that might be useful but I think that might might be important building blocks to actually building a workable computational model I do think simulating the wave could be interesting to from a different direction I was another thought that you produced was from nature inspired

metah heuristic computation or optimization where we model things like bumblebees and beehives and ants and ant colonies and I know Daniel has done work with ants as well and likes them quite a bit um and also we've done things with bats echolocation and wave transmission so if we model mathematical waves at least we could use that or Frameworks for metah heuristic computation and sort of backwards build that let's say derive the computational mechanisms that could allow us to couple neurons to use that information and then we could do actual simulations of different ways of different properties we can model the frequencies we can kind of control those hyper parameters and then in doing so we could then look at what Carl was then bringing up which is the one of the key Essences of mortal computation which is that spatial Arrangement that anatomical organization of uh neurons and their particular components um then you could also do tests like okay if we build them in a spherical kind of construct or we build them in such an anatomical construct that um is conducive to those simulated waves then you could do what Carl said computationally which is we could change the the shape of these anatom these computational Thalamus models right we can make them cubic or you know other types of weird shapes and see if squashing them changes for example how these neurons respond on simple cognitive tasks that we can emulate again we could use very simple spatial cognition tasks that we could simulate and then we could start manipulating the actual uh virtual morphology as Coler would call it or just the virtual anatomy and then that could give you some at least computational tests of the effect in value of wave computation um I think that's the key and I'm not even sure if what I just said mostly is rubbish and uh just me kind of generating what I would do if I had to build something like that and I think that might be one of the single most important questions because I think it answers both neuron models without waves and neurons plus waves as you said we need to build both those models and I think that kind of coupling and that modeling of that relationship is a first step and then of course then you could then start to look at actual physical constructs that relate that computation kind of like a reverse we're using computation to drive what we think the structure should be and then say are there actual structures that we find and then we can constrain our models a little bit better as we do have anatomic or we do have computational models of phallic structures like Chris alus group has a phalus structure in their nango framework uh again spawn I think has like a Thalamus Arch component um so the and those are very biophysically grounded like they use anatomical values things that are actually grounded from neurophysiological data so that could be an interesting starting point and that could allow us to then actually test and uh simulate what Carl was saying about

squishing and stretching and uh changing the form of the thalamus so that's something I thought might be worth mentioning and that's where the Mortal computation this really comes in play because again it's that specific morphological Arrangement that actual construction and relationship between your owns that's really important and changing that would change the computation that's what another byproduct of our framework was basically saying is that you're not going to get the same computation if you change the anatomy if you change this um so that's that's an important part and then just quickly I wanted your last slide before this one I don't know how to pronounce the name of that uh that entity either um but I thought that was really interesting because if you were saying it has essentially like a little eye but it doesn't have any known neurons or anything that we can perceive directly so could that be neurons without cognition and I think that gets at to what uh uh Michael Levan and again Carl and I site and talk with him a lot which is the idea of Basil cognition and I think that's kind of where we don't really have that quite right from a computational point of view so I do think that's even more important to try to model these uh Elementary cognitive systems and again looking at these building computational models I think of like that single cell organism could be valuable and maybe give us some insights as to well okay what is it that we need to get this thing to actually solve very very simple tests like follow gradients of chemicals which we do that in chemotaxis and phototaxis experiments and we do build little models uh of for example E. coli trying to Traverse these little uh little niches and I think that's the other part that you're going to need in building uh even simulated versions of what you seek which is you're going to need that virtual Niche because again and you point out Robert which I was happy to see that that you know your what you're thinking is that it's Elemental embodied but I think the other part that can't be ignored if you're going to properly uh examine and simulate mortal Computing systems is you need a niche need that more uh virtual environment to actually and those physical properties need to be very explicit um when you build it so I think that that was something that I think is going to be also very difficult uh but probably the most important to make the next step for Mortal computation and bi intelligence hopefully some of that made sense no thanks Alex I I agree with you that the simple animals the simple comp simple brains are the place to look I I recall that quantum mechanics started with the hydrogen atom not with the uranium atom I'll I'll actually kind of add a complimentary view I was wondering to what extent will the kind of representationalist form and context for the wave that you described in humans be of maybe even a slightly different kind in a single cellular or in an insect so for example there's the outfielder catching a Fly ball who yes you

might be able to describe that in terms of the spatial temporal frequency of the baseball moving through space and the body moving through space there also may be a horizon such as following a gradient of shade that also gives a behavioral solution or even an indistinguishable approach so that that may speak to again the variety of possible substrates of mortal computers because the insect's body and this is the to the empirical Point how can we differentiate the situation where spatial representation is happening with or without a causality closure from where behavioral strategies are integrated like moving away from a shadow for a fly it may not require the spatial Precision on the object so actually just determining where and how variation across different species and with across individuals and settings how would we differentiate that from a null hypothesis where neural or wave wave particle or both or neither we don't even need to invoke space we just invoke reflexive Behavior I agree that's a very important question it's addressed a little bit in the paper but what we need is tests of insect behavior for instance which depend on the 3D spatial model which don't depend on just as you say detecting a shadow or detecting a looming thing uh and it's quite difficult to devise those tests yeah I'll read a question from the live chat and anyone can give a thought okay Courtney writes thank you for the interesting talk could this Theory be used to test whether or not someone in the ICU with brain damage is conscious and if so to what degree could it be tested similarly to integrated information Theory or are we far away from something like that I'd say we're far away because even experimentally finding the wave is far away we have to work out what the biophysics of it is before we can build detectors for it and that may be as far away as as detection of dark matter or for the neutrino for instance that was theoretically predicted it took 25 years to find it experiment I think it may take a long time to find a wave experimentally so in terms of measures of Consciousness I'd say we're a long way off thanks um Ken deid any please go for it well I just wanted to mention one thing um that this whole discussion reminded me of a long time ago I think I was in graduate school I read a book by Ralph D Ellis not to be confused with the George Ellis that um that Carl mentioned but it's called a book called an ontology of Consciousness and if my memory serves it defends just the general idea that Consciousness is a wave um and defends the kind of wave medium model for thinking about the Mind Body problem and I I just remember it being very interesting but I don't but that's so long ago that you might read it I might read it now and be totally disappointed but just something to throw out there send me the reference that be great yeah yeah Ralph DL I can send you that an ontology of Consciousness but I do remember being being struck by it as being sort of plausible but you know how those things go it's like

it's metaphysics it's ontology you know yeah you're pretty far away from physical testing but in terms of some of the sort of theoretical advantages of thinking about it that way it might be worth a look I'll put it in the chat thank you de any thoughts and then anyone watching can write any last questions and then otherwise Robert let's hear like what are your next moves and just kind of summarize the commentary affordance maybe I I can you know I I this is a kind of question that you know anybody interested in topic struggle with in kind of okay what is the sort of Integrative mechanism uh that is required to understand things like Consciousness uh or does it relate to the brain with all the limitation of both you know theoretically and methodologically of of contemporary Neuroscience uh I'm totally open now at this point to the possibilities that yeah there is you know some form of physics that we haven't understood about the brain or that is related to the brain or interact with the brain and becoming like even open to some form of you know physical dualism I don't know you would call that uh I'm sure that it's it's going to be very hard to progress with that that we are still at a sort of metaphysical age of this type of Science and discoveries uh we don't have the principle yet maybe uh I doubt but I'm not a physicist so it's really like you know talking that Einstein condensate will address the question I mean think about the the size that has been achieve the maximal size of both Einstein condensate and under what circumstances irrespective of you know okay uh temperature issue you know but it's what like a picol lit like billion atoms um so you would have like billions and billions of bosen shin condensate that would have I mean it's it seems as I I don't I I don't have a good gut feeling about it but uh there is also more mundane things like you know we had this discussion about uh electromagnetic wave in the brain at the kind of micro scale or mesoscale as EP phenomenon but as I said before although I don't recall the names of the authors working on that but there is a bunch of labs and researcher working on that uh there is now or at least there's some theoretical well formulated ated theoretical ground and some empirical evidence for inductive mechanism in the brain basically you have you know currents going through uh um your axon bundles of fibers uh you know your action potential if you wish that are associated with those currents that induce electromagnetic wave that propagate and may have a role of for instance synchronization with another bundle just next to it so are those things are very minute as effects apparently uh they don't have a very very large long range uh um effect that's because of biophysical constraint if I recall well but it talks about the possibility that you know we may not want to throw the baby out of the bath water classical electromagnetism may still have something to say and in particular if this kind of induction mechanism uh which is literally top down I mean the

downward cation if you wish you know there is a fiber bundle that generate electromagnetic waves that then affect currents in another fiber bundle we could say well you know then you could imagine something quite sophisticated and in which the role of anatomy the grow Anatomy the shape of the brain the organization then might become uh very important to understand because of course the relationships if those electromagnetic wave induction are short range you know how you bring anatomically things closer in order to make them interact becomes very important and maybe that connects back to your thalamic hypothesis but again what I'm talking about is very very very speculative thanks yeah I agree um my my feeling is the biophysics is wide open so electromagnetism May well be a part of it and you used Mars levels earlier it's possible the wave is has different implementations like I think this conversation brought up a lot of different dimensions and also even begets bigger questions like do scientific hypotheses have for and against is there only one hypothesis chosen or could there be functionality distributed across multiple biophysical systems such that something being a AR true or false which is kind of predicated the whole basian epistemology that doesn't necessarily need to be true there could be utility continua for a theory and a predictive continua yeah I I guess the hint is that since there's ever a hint of the wave in both the thalamus and insects and opposite ends of the animal tree evolutionary tree that the wave hypothesis might carry across the whole lot and in which case you ask is it convergent evolution and different species different branches or did it go right back to the pre primordial Branch before the Cambrian era I would like it to go right back yeah in that case you can consider like arthropods or look how far back but insects are only one kind of arthropod and they have yeah enough homology that you might be able to extend this argument and do ancestral State reconstruction on neuroanatomy at different levels and these analyses may have already been done and then the allometry of those reconstructions can be compared to yeah any last comments or um Robert with last word yeah well many thanks to all of you for for your comments um it's been most useful and I guess I just finally request if I I would like to have to get some commentary from all sorts of people on this hypothesis so if you would send me some comments just initial ones and gut reaction comments if you like uh on this core question of can we dismiss the wave hypothesis or do we need to take it seriously then that would uh kind of kick off some of the commentary anyway many thanks thank you to all the guests and thank you Robert all right farewell thank you bye everyone bye thank you